

➤ Project Proposal: Multi-Tenant ERP & Mobility Platform

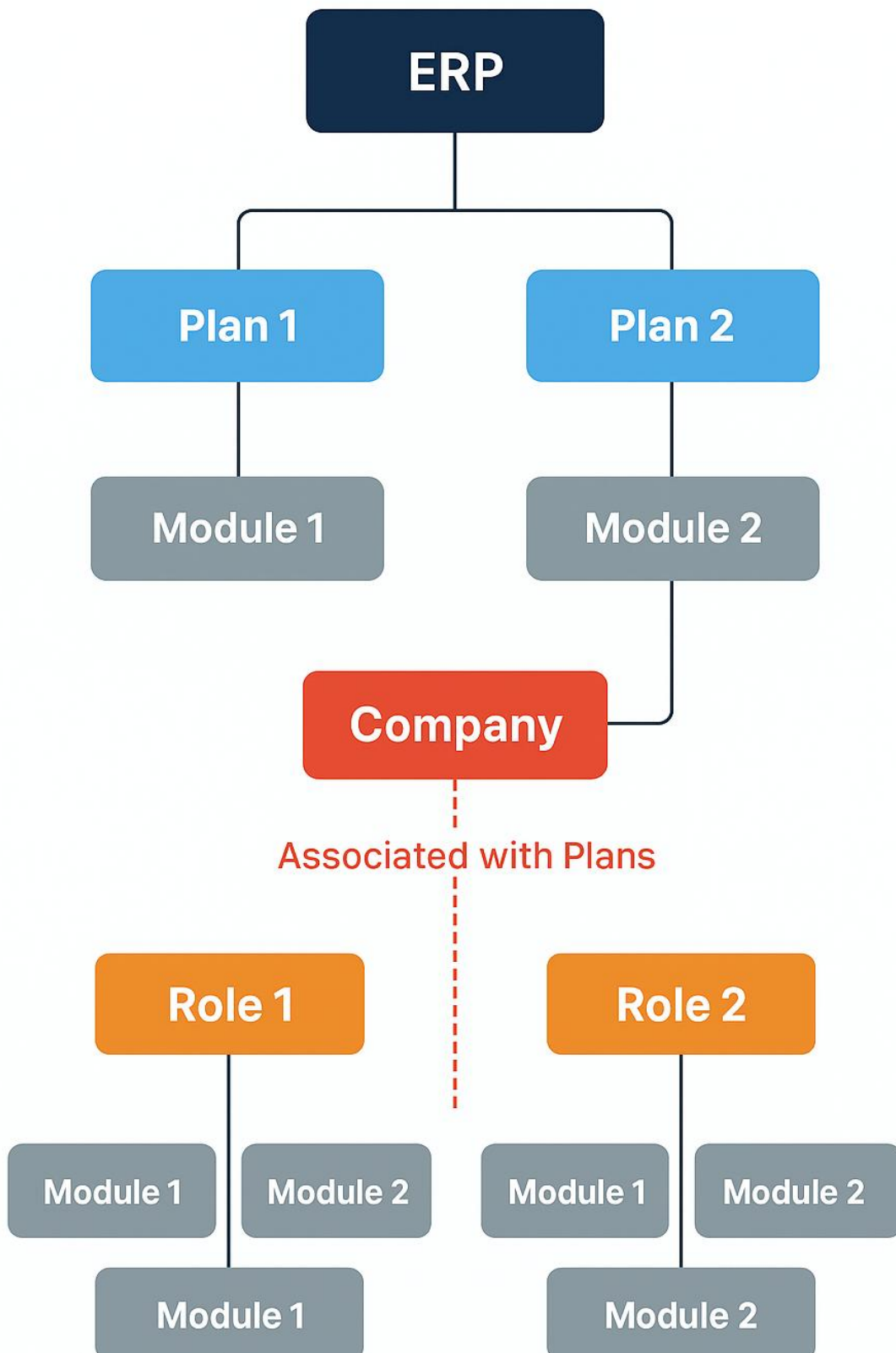
1. Introduction

This proposal outlines the design and development of a multi-tenant ERP platform aimed at simplifying and digitizing essential business operations such as visit management, lead management, order processing, trip planning, stock transactions, and service workflows. The solution is built to unify field teams and office teams through a modern web portal and a dedicated mobile application, enabling real-time updates, structured collaboration, and improved process visibility.

The ERP will follow a modular, plan-based architecture where companies (tenants) subscribe to specific plans, and each plan provides access to selected modules. Within each company, administrators can create user roles with tailored permissions, ensuring that every user has access only to the modules and features relevant to their responsibilities.

Beyond streamlining daily tasks, this ERP is designed to improve accountability, reduce manual dependencies, and provide decision-makers with accurate insights through dashboards and reports. The platform is crafted with scalability in mind, allowing it to support growing teams, expanding operations, and the onboarding of multiple organizations on a single, secure infrastructure.

Overall, this project aims to deliver a practical, reliable, and enterprise-ready ERP system that brings structure, transparency, and efficiency to business operations—making it easier for companies to manage their teams, customers, and workflows from one unified ecosystem.



2. System Architecture Overview

The ERP is designed around a centrally managed multi-plan structure:

- **ERP Core**

The main platform hosts multiple subscription plans.

- **Plans & Modules**

Each plan includes a set of modules (Visit, Lead, Order, Service, Stock, etc.).

Companies subscribe to plans based on their operational needs.

- **Tenant Companies**

A company is associated with one or more plans. These mappings determine which modules are available to its users.

- **Roles & Permissions**

Each company configures roles (e.g., Sales Executive, Manager, Service Engineer).

Roles dictate module-wise access.

This layered structure ensures flexibility, scalability, and clear separation between tenants.

3. Technology Stack

To deliver a stable, scalable platform, the following technologies will be used:

Frontend (Web Portal)

- **React.js**
- Component-based UI architecture

- Responsive layout for all admin & dashboard views
- Secure API communication via JWT or OAuth

Python (FastAPI / Django Rest Framework)

- Multi-tenant architecture
- High-performance RESTful API development
- Excellent support for async operations (FastAPI)
- Built-in security, validations, and clean architecture
- Background task processing (Celery + Redis)

Database

- **Microsoft SQL Server**
- Normalized schema for ERP modules
- Company-wise data separation
- Optimized indexing for reporting and dashboards

Mobile Application

React Native

- Offline-first capability for field teams
 - Real-time sync with backend
 - Push notifications for assignments, leads, and service updates
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4. Major Functional Modules

1. Visit Management

- Schedule and track customer visits
- Field team check-in/check-out using mobile app
- Route planning and trip linkage
- Real-time location logs (optional geo tagging)

2. Lead Management

- Capture leads from field, web, or import sources
- Pipeline stages (New → Qualified → Quoted → Closed)
- Assign leads to team members based on role
- Automated follow-up reminders

3. Lead Tracking

- Timeline view of all interactions
- Notes, communication history, and updates
- Performance tracking for sales teams

4. Trip Management

- Trip planning for field teams
- Multi-location routes
- Approval workflow for planned trips
- Expense & distance logging

5. Quotation Module

- Create and share quotations directly from the platform

- Predefined templates per company
- Version control of quotes
- Approval hierarchy for high-value quotes

6. Order Management

- Convert quotations into orders
- Order assignment and status flow (Pending → Processing → Dispatched → Completed)
- Customer communication through SMS/Email

7. Service Call Management

- Track customer service requests
- Ticket creation, assignment, and resolution
- Spare parts usage logging
- SLA monitoring and escalation alerts

8. Customer Communication

- Unified module for automated and manual communication
- SMS, Email, WhatsApp integration
- Templates based on company standards

9. Reporting & Dashboards

- Real-time dashboards for leads, sales, visits, service, orders, and stock
- Role-based report visibility
- Downloadable reports (Excel, PDF)

10. Stock Management

Includes:

- **a. Stock Receipt** – GRN, material inward log
 - **b. Warehouse Transfer** – stock movement from one location to another
 - **c. Sample Request** – request/approve/issue samples to field teams
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5. Multi-Tenant Architecture Approach

- Single codebase serving multiple companies
- Company-specific configurations: roles, users, modules, branding
- Logical data isolation through tenant IDs
- Plan-based module activation per tenant
- Scalable deployment with load balancers and caching

This approach ensures efficient onboarding and maintenance without data leakage between tenants.

6. Security & Access Control

- JWT-based authentication for both web and mobile
- Role-based access control mapped at module and feature level

- Encrypted data storage for sensitive fields
 - Optional two-factor authentication for admins
 - Audit logs for all major activities
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7. Deployment & Scalability

- Cloud-based deployment (Azure/AWS)
 - Auto-scaling for peak loads
 - CDN for static assets
 - Queue workers for time-consuming tasks
 - API rate limiting for security
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8. Expected Outcomes

This ERP will enable companies to:

- Streamline field operations
- Improve lead conversion rates
- Enhance service performance
- Reduce manual errors
- Strengthen customer communication

- Gain visibility with real-time dashboards
- Manage stock and inter-warehouse movement efficiently

9.Team Breakdown :

- **Leadership (Support Roles – Not Counted in Main Team)**

Role	Count	Responsibility
Solution Architect	1	System architecture, module design, DevOps (CI/CD + deployments), security, performance optimization.
Technical Lead	1	Daily technical supervision, code reviews, sprint planning, ensuring best development practices.

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- **Core Working Team (5 Members)**

Role	Count	Key Responsibilities
Python Backend Developers	2	Multi-tenant setup, core backend modules, APIs for Leads, Visits, Trips, Orders, Stock, Service Calls, etc.
React Frontend Developer	1	Dashboard UI, reporting screens, module interfaces, responsive design.
Full-Stack / Frontend + Python Mix	1	Supports both frontend and backend workloads, ensures seamless integration.
QA Engineer	1	Functional testing, regression, bug tracking, UAT validation, quality assurance from Day 1.

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Total Working Team Size: 5

Total Including Leadership Support: 7

10. Project Timeline & Resource Estimate:

Phase	Estimated Duration	Details
1. Discovery & Architecture	2–3 weeks	Requirement detailing, technical planning, multi-tenant structure design, and architecture validation.
2. Backend Core Setup	3–4 weeks	Parallel development of authentication, tenant layer, user/role/permissions, plan–module mapping, and base APIs.
3. Module Development	8–10 weeks	Larger backend team enables simultaneous development of Visit, Lead, Tracking, Trip, Quotation, Orders, Service Calls, and Stock Management modules.
4. Frontend Web Portal	6–8 weeks	Two React developers handle dashboards, reporting, module UI, and company settings concurrently.
5. Mobile App (React Native)	8–10 weeks	Mobile features (Visit, Tracking, Notifications, Offline Sync) built in parallel with backend APIs.
6. QA, UAT & Hardening	3–4 weeks	Continuous testing throughout development ensures stable UAT and faster final validation.

7. Deployment & Training	1–2 weeks	Cloud deployment, CI/CD setup, documentation, and tenant onboarding.
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→ Total Estimated Duration: 18–20 Weeks (4–5 Months)

11. Comparable ERP Models & Industry References

Our proposed ERP system follows best practices used by well-known ERP solutions across the world. This helps ensure that the platform is modern, scalable, and aligned with industry standards.

Similar ERP models include:

- [Zoho ERP Suite](#) – Modular ERP with CRM, Inventory, Quotations, and Orders.
- [Odoo ERP](#) – Flexible modular structure used for sales, inventory, and service workflows.
- [SAP Business One](#) – Multi-module ERP for SMEs covering sales, service, and stock operations.
- [Microsoft Dynamics 365](#) – Cloud-based ERP with strong multi-tenant capabilities.
- [Freshworks CRM + Field Service](#) – Visit management, service calls, and customer workflows.
- [Dolibarr ERP & CRM](#) – Lightweight open-source ERP/CRM, suitable for small businesses needing basic ERP functionality.

- [ERPNext](#) – Open-source ERP for SMEs: accounting, CRM, inventory, HR, projects, manufacturing, and more.
- [LedgerSMB](#) – ERP/accounting system with support for small and medium businesses, focusing on accounting, billing, financials, and basic ERP needs.
- [webERP](#) – Web-based ERP focused on inventory, financials, distribution — good for small businesses with straightforward needs. web-erp.org (or search *webERP*)

Concepts we adopt from these systems:

- A single platform to manage the full customer journey (Lead → Quotation → Order → Service).
- Dashboards that change based on user role.
- Modules that can be turned on/off per tenant.
- Mobile-first approach for field teams (visit logging, tracking, service calls, etc.).

This ensures our ERP follows the standards of globally successful platforms.

12. Unique Value Proposition

Key strengths that make this ERP different:

- Modules can be activated or deactivated for each tenant.
- Complete automation from lead creation to order and service handling.
- Mobile app supports offline mode and GPS for field visits.

- Dashboards and reports are customizable per tenant.
 - Cloud-based multi-tenant setup that scales easily.
 - Faster implementation time compared to traditional ERPs.
 - More cost-effective than SAP, Dynamics, or Freshworks for SMEs.
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13. Technology Advantages

Why the chosen tech stack is suitable:

- **React.js:** Fast and modern UI for the web portal.
 - **Python (FastAPI/Django)** for scalable & secure backend
 - **React Native:** One codebase for both Android and iOS apps.
 - **SQL Server:** Reliable, scalable database for structured business data.
 - Architecture supports microservices for future expansion.
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14. Security & Compliance

Security is a core part of the system design:

- Data is separated for each tenant to avoid cross-access.
- Role-based permissions control what each user can see or do.

- Encrypted login and secure API communication.
 - Logs maintained for important user actions.
 - Full backup and disaster recovery plan.
 - System structure ready for GDPR-style data protection.
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15. Deployment Architecture

How the system will run in production:

- Hosted on cloud platforms like AWS, Azure, or Google Cloud.
 - CI/CD pipeline for smooth code deployment.
 - Auto-scaling to support growing tenants.
 - Centralized logging and monitoring for performance and errors.
 - API gateway used for managing integrations.
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16. Risks & Mitigation Plan

Key Risks

- Multiple modules need to work together smoothly.
- Heavy data usage by field teams using the mobile app.
- Maintaining strict data separation for tenants.

Mitigation Strategy

- Strong architectural oversight by the senior architect.
 - Independent modules connected through clean APIs.
 - QA begins early to reduce end-stage problems.
 - Performance and load testing before go-live.
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17. Support, Maintenance & SLA

What clients receive after the system goes live:

- 24×7 support for critical issues.
- Regular bug fixes, stability updates, and improvements.
- Enhancements based on client feedback.
- Performance and usage monitoring.
- Monthly maintenance and system health reports.
- Dedicated account manager for enterprise-level tenants.